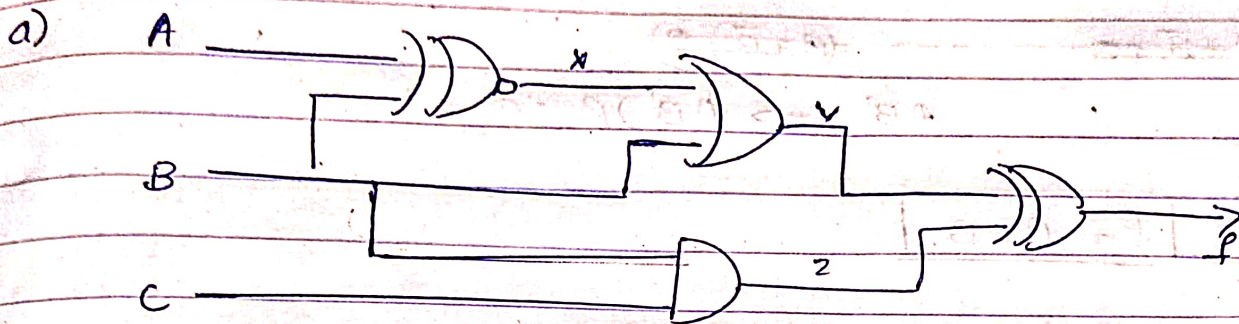


Day 2: Assignment

1. Derive the output expression 'f' for the below circuits and draw the simplified circuits.



$$A \oplus B' = AB + A'B' = X$$

$$Y = X + B$$

$$= AB + A'B' + B$$

$$AB + B = B$$

$$Z = BC$$

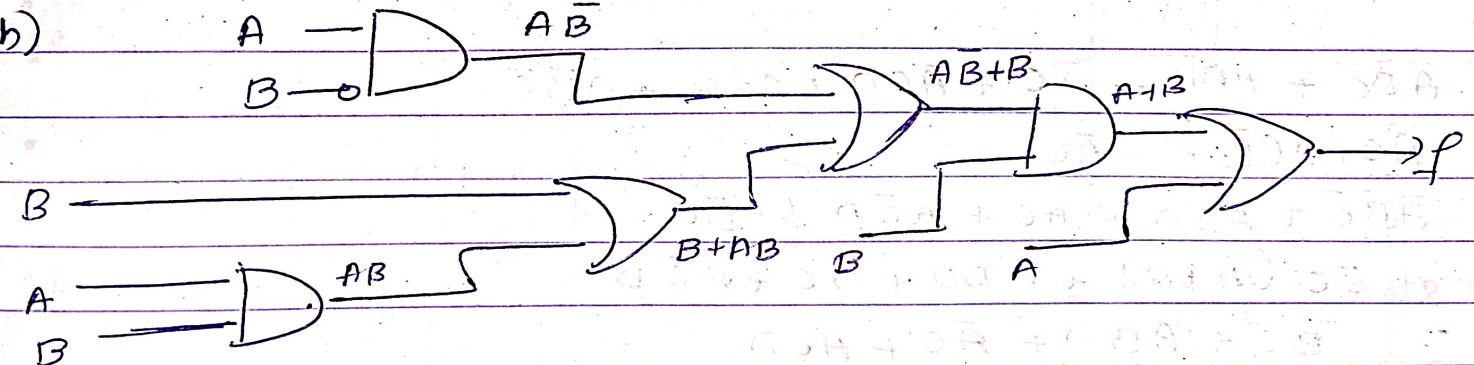
$$Y = B + A'B'$$

$$= B + A'B' = (B + A')(B + B') = B + A'$$

$$Y = A' + B$$

$$f = (A' + B) \oplus BC$$

$$= A'B' + A'C' + BC'$$



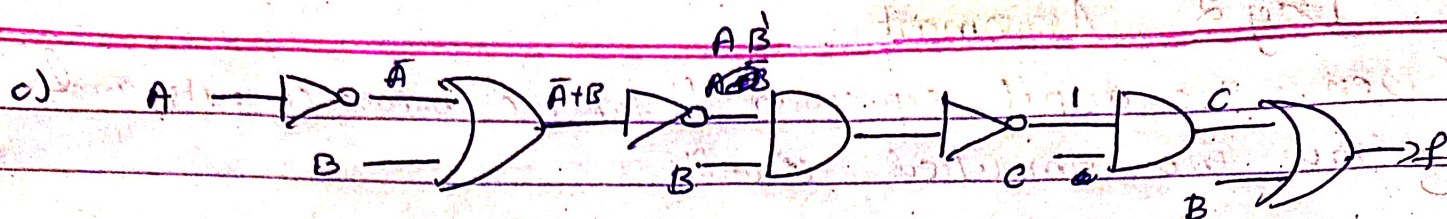
$$B \quad A\bar{B} + B + AB$$

$$A\bar{B} + B$$

$$B + A \quad B(A\bar{B} + B) = A + B$$

$$A + A + B$$

$$f = A + B$$



$$\overline{A+B} \cdot B$$

$$AB' \Rightarrow (AB')B = 0 = 1 \rightarrow$$

$$P = C + B$$

2) Minimise the following Boolean Expressions using Boolean postulates.

$$\begin{aligned} \text{a) } & \overline{A}C\overline{D}E + \overline{A}B\overline{D} + ABCE + ABD \\ &= \overline{A}C\overline{D}E + \overline{A}B\overline{D} + AB(CE + D) \\ &= \overline{A}D(CE + \overline{B}) + AB(CE + D) \\ &= CE(\overline{A}D + AB) + \overline{A}B\overline{D} + ABD \\ &= CE(AB + \overline{B}D + \overline{A}D) + \overline{A}B\overline{D} + ABD \\ &= ABCE + \overline{B}DCE + \overline{A}DCE + \overline{A}B\overline{D} + ABD \\ &= AB(CE + D) + \overline{B}DCE + \overline{A}DCE + \overline{A}B\overline{D} \\ &= \underline{\underline{ABD + BCE\overline{D} + \overline{A}B\overline{D}}} \end{aligned}$$

$$\text{b) } \overline{A}\overline{B}\overline{C} + ABD + \overline{A}C + \overline{A}C\overline{D} + A\overline{C}D + A\overline{B}\overline{C} \\ \overline{A}C + \overline{A}C\overline{D} \rightarrow \overline{A}C$$

$$\begin{aligned} &= \overline{A}\overline{B}\overline{C} + ABD + \overline{A}C + \overline{A}C\overline{D} + A\overline{B}\overline{C} \\ &> \overline{B}\overline{C}(\overline{A} + A) + ABD + \overline{A}C + \overline{A}C\overline{D} \\ &> \overline{B}\overline{C} + ABD + \overline{A}C + \overline{A}C\overline{D} \\ &> \overline{B}\overline{C} + \overline{A}C + AD(\overline{B} + \overline{C}) \\ &> \overline{B}\overline{C} + \overline{A}C + A\overline{C}D + ABD \\ &> \underline{\underline{\overline{B}\overline{C} + \overline{A}C + AD(\overline{C} + B)}} \end{aligned}$$

$$c) \bar{A}\bar{C} + BC + \bar{A}\bar{B} + \bar{A}BC + AC\bar{D} + \bar{B}\bar{C}\bar{D}$$

~~$$\bar{A}\bar{C} + BC + \bar{A}\bar{B}\bar{C}$$~~

$$\bar{A}\bar{C} + BC(\bar{A}+1) + \bar{A}\bar{B}\bar{C} \cdot AC\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$\bar{A}\bar{C} + BC + AC\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$(AC + \bar{A}\bar{C})\bar{D} + (BC + \bar{B}\bar{C})\bar{D}$$

$$\bar{A}\bar{C} + BC + \bar{A}\bar{B}$$

$$d) \bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_1x_2\bar{x}_3x_4 + x_2$$

$$\bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + x_1x_2\bar{x}_3x_4 + \bar{x}_1x_4(x_2 + \bar{x}_2)$$

$$\bar{x}_4 + \bar{x}_1x_4 + x_3x_4 + x_1x_2\bar{x}_3x_4$$

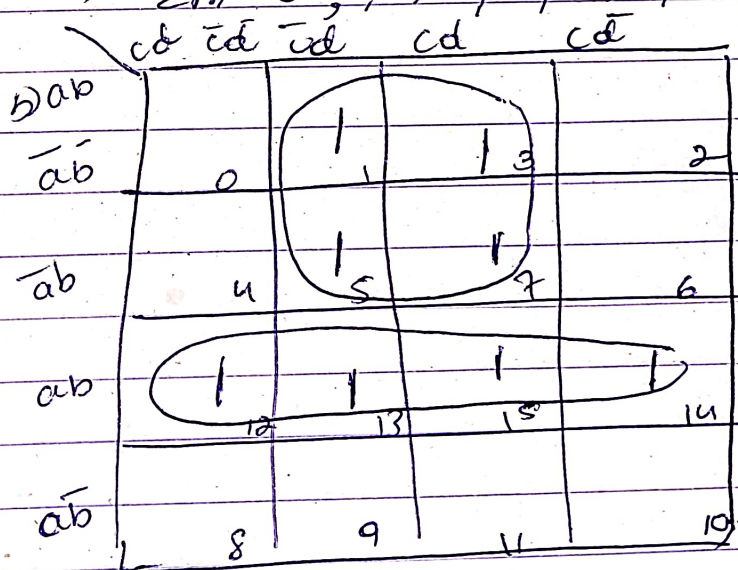
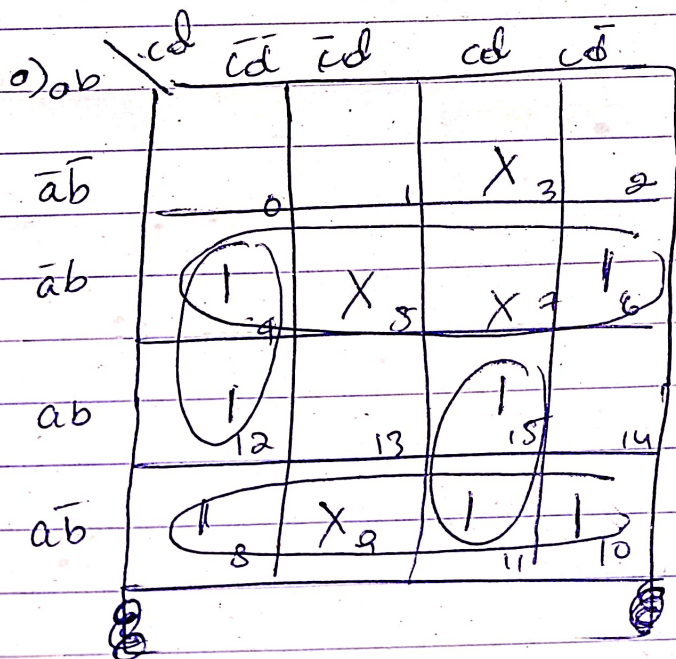
$$\bar{x}_1 + x_3x_4 + \bar{x}_1\bar{x}_3$$

4) Minimise the following Boolean Expression using Kmap

$$a) f(a,b,c,d) = \sum_m (4,6,8,10,11,12,15) + \sum_d (3,5,7,9)$$

$$b) f(a,b,c,d) = \prod_m (0,2,4,6,7,9) \oplus \prod_d (10,11)$$

$$\hookrightarrow \sum_m (1,3,5,8,12,13,14,15)$$



$$\bar{a}b + \bar{a}d$$

$$\bar{a}b + \bar{a}\bar{b} + b\bar{c}\bar{d} + a\bar{c}d$$